

4.2

We approximate this binomial distribution ($X \sim \text{Bin}(n = 1000, p = 0.42)$) because $np(1-p) > 10$.

$$E(X) = np = 420$$

$$\text{Var}(x) = np(1-p) = 159.6$$

$$X \sim N(420, 159.6)$$

$$\sigma = \sqrt{159.6} = 12.633$$

$$450 = 420 + 12.633z \rightarrow 30 = 12.633z \rightarrow z = 2.37$$

$$P(X > 450) = 1 - P(X < 450) = 1 - \Phi(2.37) = 1 - 0.9972 = 0.0028$$

4.6

$$2\Phi(2\epsilon\sqrt{n}) \geq 0.95 \rightarrow \Phi(2\epsilon\sqrt{n}) \geq 0.975$$

$$\rightarrow 2\epsilon\sqrt{n} \geq 1.96 \rightarrow n = \frac{1.96^2}{2\epsilon}$$

$$\epsilon = 0.02 \text{ from the problem}$$

$$n = \frac{1.96^2}{0.04} = 2401$$

4.10

$$1 - P(X = 0) = 0.5 \rightarrow P(X = 0) = 0.5$$

$$P(X = 0) = e^{-\lambda} * 1 = 0.5 \rightarrow \lambda = -\ln(0.5) = 0.693147$$

$$\text{Scoring exactly 3 goals (hat trick) is then } P(X = 3) = 0.5 \frac{3^{-\ln(0.5)}}{3!} = 0.178457$$

If you define hat trick as 3 or more goals then its $1 - P(X = 0) - P(X = 1) - P(X = 2)$ but since the majority accepted definition is exactly 3 goals I will leave my answer as $P(\text{HATTRICK}) = P(X = 3) = 0.178457$

4.16

a

$$P(Start1) = \frac{1.99999999-1.5}{4.8-1.5} \approx \frac{0.5}{3.3} = \frac{5}{33}$$

$$X \sim Bin(500, \frac{5}{33})$$

$$X \sim N(\frac{2500}{33}, \frac{70000}{1089})$$

$$\frac{2500}{33} + \sqrt{\frac{70000}{1089}}z = 65$$

$$z = \frac{-355}{\sqrt{70000}} = -1.34$$

$$P(X < 65) = \Phi(z) = 0.09012$$

b

$$P(Start3) = \frac{3.99999999-3}{4.8-1.5} \approx \frac{1}{3.3} = \frac{10}{33}$$

$$X \sim Bin(500, \frac{10}{33})$$

$$X \sim N(\frac{5000}{33}, \frac{115000}{1089})$$

$$\frac{5000}{33} + \sqrt{\frac{115000}{1089}}z = 160$$

$$z = \frac{280}{\sqrt{115000}} = 0.826$$

$$P(X > 160) = 1 - \Phi(z) = -0.20327$$

4.24

a

Let $\#4_n$ denote the number of 4's in n rolls.

$$P(\frac{\#4_n}{n} \geq 0.17) = P(\frac{\#4_n}{n} - \frac{1}{6} \geq 0.17 - \frac{1}{6})$$

$$\leq P(|\frac{\#4_n}{n} - \frac{1}{6}| \geq 0.17 - \frac{1}{6})$$

$$= 1 - P(|\frac{\#4_n}{n} - \frac{1}{6}| < 0.17 - \frac{1}{6}) = 0 \text{ as } n \rightarrow \infty$$

b

Let $\#S_n$ denote the set $[\#1_n, \#2_n, \#3_n, \#4_n, \#5_n, \#6_n]$

$P(A_n) = \forall S_n \in \#S_n$ where $P(0.16 < \frac{S_n}{n} < 0.17)$

$$\begin{aligned} \text{Due to normal approximation: } P(0.16 < \frac{S_n}{n} < 0.17) &= \Phi((0.17 - \frac{S_n}{n})(\frac{\sqrt{n}}{\sqrt{\frac{1}{6} \frac{5}{6}}})) - \Phi((0.16 - \frac{S_n}{n})(\frac{\sqrt{n}}{\sqrt{\frac{1}{6} \frac{5}{6}}})) \\ &\approx \Phi((0.17 - \frac{1}{6})(\frac{\sqrt{n}}{\sqrt{\frac{1}{6} \frac{5}{6}}})) - \Phi((0.16 - \frac{1}{6})(\frac{\sqrt{n}}{\sqrt{\frac{1}{6} \frac{5}{6}}})) \\ &= \Phi(0.008894427\sqrt{n}) - \Phi(-0.01788854\sqrt{n}) \end{aligned}$$

We want $\Phi(0.008894427\sqrt{n}) - \Phi(-0.01788854\sqrt{n}) > 0.999$ so pick a large $n = 160000$ and solve we get $\Phi(3.578) - \Phi(-7.155) \approx 1$ which is greater than 0.999.

4.26

$$P(|\hat{p} - p| < 0.1) \geq 2\Phi(2(0.1)\sqrt{n}) - 1 = 2\Phi(2) - 1 = 2(0.97725) - 1 = 0.9545$$

With confidence of 0.9545

4.35

a

Since $X \sim \text{Bin}(365, \frac{1}{2^9})$ (note its 2^9 because the first flip can be either heads or tails but all subsequent flips must be same as first flip).

$$P(X > 1) = 1 - P(X = 1) - P(X = 0)$$

$$P(X = 0) = (1 - \frac{1}{2^9})^{365}$$

$$P(X = 1) = 365(1 - \frac{1}{2^9})^{364} \frac{1}{2^9}$$

$$P(X > 1) = 1 - 365(1 - \frac{1}{2^9})^{364} \frac{1}{2^9} - (1 - \frac{1}{2^9})^{365}$$

b

Since $X \sim \text{Bin}(365, \frac{1}{29})$ (note its 2^9 because the first flip can be either heads or tails but all subsequent flips must be same as first flip).

This would not be a good fit for a normal approximation because p is close to 0 and $np(1-p) \approx 0.711 < 10$.

Thus $\lambda = \frac{365}{29} = 0.71289$

$$P(X > 1) = 1 - P(X = 1) - P(X = 0)$$

$$P(X = 0) \approx e^{-0.71289}$$

$$P(X = 1) \approx e^{-0.71289} \frac{0.71289}{1}$$

$$P(X > 1) = 1 - 1.71289e^{-0.71289} = 1 - 0.8397 = 0.1603$$